Q-FW (Yurtsever et al., 2022): Sections 3–4 Extraction and Comparison

2026-03-09

1 Scope

This note summarizes the method in Sections 3 and 4 of: *Q-FW: A Hybrid Classical-Quantum Frank-Wolfe for Quadratic Binary Optimization.* It then compares that method against the current implementation in this repository.

2 Section 3: Problem Formulation (Extracted Method)

The paper starts from a constrained quadratic binary optimization (QBO):

$$\min_{x\in\{0,1\}^n} \; x^\top Qx + 2s^\top x \quad \text{s.t.} \quad a_i^\top x = b_i,\; i = 1,\dots,m. \tag{1}$$

Main reformulation steps:

1. Symmetrize Q without changing the objective.
2. Encode linear term into quadratic form for binary x.
3. Lift to matrix variable $X = xx^\top$ (rank-1, completely positive).
4. Rewrite objective and constraints linearly in (x, X):
   - $x^\top Qx = \mathrm{Tr}(QX)$,
   - $(a_i^\top x)^2 = b_i^2 \iff \mathrm{Tr}(A_iX) = b_i^2$, where $A_i = a_ia_i^\top$.
5. Replace nonconvex constraints with complete positivity constraints using the lifted matrix $W = \begin{bmatrix} 1 & x^\top \\ x & X \end{bmatrix} \in \Delta^p$, with $p = n + 1$.
6. Compact form:
$$\min_{W\in\Delta^p} \; \mathrm{Tr}(CW) \quad \text{s.t.} \quad \mathcal{A}W = v, \tag{2}$$
where $\mathcal{A}$ and v collect all affine constraints, including $\{a_i^\top x = b_i\}$, $\{\mathrm{Tr}(A_iX) = b_i^2\}$, $\mathrm{diag}(X) = x$, and $W_{1,1} = 1$.

This CP model is convex but NP-hard because of complete positivity.

3 Section 4: Quantum Frank-Wolfe (FWAL/FWQP)

The algorithm solves the CP model via an augmented Lagrangian:

$$\mathcal{L}_\beta(W; y) = \mathrm{Tr}(CW) + y^\top(\mathcal{A}W - v) + \frac{\beta}{2}\|\mathcal{A}W - v\|_2^2, \tag{3}$$

with primal variable $W \in \Delta^p$, dual variable $y \in \mathbb{R}^d$, and penalty $\beta > 0$.

At iteration t (paper notation):

1. Increase penalty: $\beta_t = \beta_0\sqrt{t+1}$.
2. Primal gradient matrix:
$$G_t = C + \mathcal{A}^* y_t + \beta_t\, \mathcal{A}^*(\mathcal{A}W_t - v). \tag{4}$$
3. LMO over complete-positive domain:
$$H_t \in \arg\min_{W \in \Delta^p} \mathrm{Tr}(G_t W). \tag{5}$$
This is reduced to QUBO by solving
$$w_t \in \arg\min_{w \in \{0,1\}^p} w^\top G_t w, \quad H_t = w_t w_t^\top. \tag{6}$$
4. FW primal update:
$$W_{t+1} = (1 - \eta_t)W_t + \eta_t H_t, \quad \eta_t = \frac{2}{t+1}. \tag{7}$$
5. Dual ascent:
$$g_t = \mathcal{A}W_{t+1} - v, \quad y_{t+1} = y_t + \gamma_t g_t. \tag{8}$$
In practice they use constant $\gamma_t = \beta_0$.
6. FWQP is recovered by disabling dual updates ($y_0 = 0$, $\gamma_t = 0$).

Rounding in Section 4:

1. From approximate $\hat{W}$, remove first row/column to get $\hat{X}$.
2. Compute best rank-1 approximation $\hat{x}\hat{x}^\top$ of $\hat{X}$ (Frobenius norm).
3. Optionally project $\hat{x}$ to feasible set of the original QBO (for permutations, Hungarian projection).

4 Differences vs Current Implementation

4.1 Summary

The current code is a standard constrained FW-like loop in variable $x \in [0,1]^n$. The paper's Q-FW is a CP-space FWAL/FWQP method in lifted variable $W \in \Delta^p$ with augmented-Lagrangian primal-dual updates. This is a structural mismatch, not a minor hyperparameter mismatch.

Paper (Sections 3–4)	Current implementation	Impact
Optimize lifted CP variable W with constraints $\mathcal{A}W = v$.	Optimizes vector x directly using Q, c from original QBO; see `qfw_solver.py` lines 121–123 and 170–194, and `gradient.py` lines 23–54.	Not the CP reformulation the paper proves/uses.
Use augmented Lagrangian $\mathcal{L}_\beta(W; y)$ with dual y and increasing β_t.	No dual variable, no β_t, no primal-dual saddle updates in `qfw_solver.py`.	Missing core FWAL/FWQP mechanism.
Primal gradient is matrix $G_t = C + \mathcal{A}^* y_t + \beta_t \mathcal{A}^*(\mathcal{A}W_t - v)$.	Uses $\nabla f(x) = (Q + Q^\top)x + c$; `gradient.py` lines 57–77.	Different objective geometry and search direction.
LMO solves $\min_{W \in \Delta^p} \mathrm{Tr}(G_t W)$ via binary QUBO in $w \in \{0,1\}^p$, then $H_t = ww^\top$.	LMO minimizes linear form $\nabla f(x)^\top s$ over sampled feasible bitstrings; `linear_minimisation.py` lines 10–14 and 142–152.	Not equivalent to the paper's CP-domain LMO.
Dual step $y_{t+1} = y_t + \gamma_t(\mathcal{A}W_{t+1} - v)$.	Absent from `qfw_solver.py`.	Cannot reproduce FWAL behavior/convergence statement in Prop. 1.
Rounding from $\hat{W} \to \hat{X} \to \hat{x}$ using best rank-1 approx and optional projection.	Threshold/greedy/randomized rounding on x; `rounding.py` lines 26–129.	Different decoding path than paper.
Constraint handling is encoded in affine map $\mathcal{A}$ and residual norm $\|\mathcal{A}W - v\|$.	Constraint treatment mostly by feasible sample filtering and heuristic rounding; also fallback may pick infeasible sample, `linear_minimisation.py` lines 154–159.	Feasibility control differs from method definition.

4.2 Concrete Mismatches

5 What To Re-Implement Later (Correct Direction)

For a faithful implementation of Sections 3–4, the future solver should:

1. Construct CP-lifted data $(C, \mathcal{A}, v)$ from the QBO instance.
2. Maintain state (W_t, y_t, β_t).
3. Compute matrix gradient G_t of the augmented Lagrangian.
4. Solve LMO in lifted domain (QUBO in w, then $H_t = ww^\top$).
5. Apply primal update for W and dual ascent for y.
6. Decode $\hat{W}$ with rank-1 approximation and projection to original feasible set.

6 Code References

The comparison above used these files:

- `quantum_optimization/pipeline/solvers/frank_wolfe/qfw_solver.py`
- `quantum_optimization/pipeline/solvers/frank_wolfe/gradient.py`

- quantum_optimization/pipeline/solvers/frank_wolfe/linear_minimisation.py
- quantum_optimization/pipeline/solvers/frank_wolfe/rounding.py
- quantum_optimization/pipeline/solvers/frank_wolfe/qfw_utils.py